

Stony Brook University
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Spring 2017

MIDTERM EXAM III

NAME: _____

ID: _____

Instructions: Answer all questions. The total number of points for the exam is 100. The number of points assigned to each section is indicated. Each question within a section carries the same weight. Calculators are allowed but other electronic devices or outside materials are NOT allowed. For questions with a numerical answer, give a numerical answer. For example

$$2 + V[X]$$

is NOT the final answer if you can compute a numerical value for $V[X]$.

You do not need to compute probabilities using any of the distributions discussed during the lectures. This is, your answer can be something like

$$1 - \Phi(1.67)$$

and you don't have to calculate that this number is actually 0.04746.

You are **NOT** allowed to leave the room until you submit your exam.

GOOD LUCK!

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Part I: Multiple Choice Questions (20 points)

1. Under imperfect multicollinearity
 - (a) the OLS estimator cannot be computed.
 - (b) two or more of the regressors are highly correlated.
 - (c) the OLS estimator is biased even in samples of $n > 100$.
 - (d) the error terms are highly, but not perfectly, correlated.
2. In multiple regression, the R^2 increases whenever a regressor is
 - (a) added unless the coefficient on the added regressor is exactly zero.
 - (b) added.
 - (c) added unless there is heteroskedasticity.
 - (d) greater than 1.96 in absolute value.
3. If you had a two regressor regression model, then omitting one variable which is relevant
 - (a) will have no effect on the coefficient of the included variable if the correlation between the excluded and the included variable is negative.
 - (b) will always bias the coefficient of the included variable upwards.
 - (c) can result in a negative value for the coefficient of the included variable, even though the coefficient will have a significant positive effect on Y if the omitted variable were included.
 - (d) makes the sum of the product between the included variable and the residuals different from 0.
4. Perfect multicollinearity
 - (a) implies that it will be difficult to estimate precisely one or more of the partial effects using the data at hand
 - (b) violates one of the four Least Squares assumptions in the multiple regression model
 - (c) suggests that a standard spreadsheet program does not have enough power to estimate the multiple regression model
 - (d) happens when you omit one dummy variable.
5. If you reject a joint null hypothesis using the F -test in a multiple hypothesis setting, then
 - (a) a series of t -tests may or may not give you the same conclusion.
 - (b) the regression is correctly specified.
 - (c) all of the individual hypotheses are always simultaneously rejected.
 - (d) the F -statistic must be negative.

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6. When there are omitted variables in the regression, which are determinants of the dependent variable, then
 - (a) you cannot measure the effect of the omitted variable, but the estimator of your included variable(s) is (are) unaffected.
 - (b) the OLS estimator is biased if the omitted variable is correlated with the included variable.
 - (c) this has no effect on the estimator of your included variable because the other variable is not included.
 - (d) this will always bias the OLS estimator of the included variable.
7. In the multiple regression model, the least squares estimator is derived by
 - (a) minimizing the sum of squared prediction mistakes.
 - (b) setting the sum of squared errors equal to zero.
 - (c) minimizing the absolute difference of the residuals.
 - (d) forcing the smallest distance between the actual and fitted values.
8. To test joint linear hypotheses in the multiple regression model, you need to
 - (a) compare the sums of squared residuals from the restricted and unrestricted model.
 - (b) use several t-statistics and perform tests using the standard normal distribution.
 - (c) use the heteroskedasticity-robust F-statistic.
 - (d) compare the adjusted R^2 for the model which imposes the restrictions, and the unrestricted model.
9. The dummy variable trap is an example of
 - (a) imperfect multicollinearity.
 - (b) something that is of theoretical interest only.
 - (c) perfect multicollinearity.
 - (d) something that happens every time you use dummy variables.
10. The overall regression F -statistic tests the null hypothesis that
 - (a) the slope coefficients are zero.
 - (b) all slope coefficients and the intercept are zero.
 - (c) the intercept in the regression and at least one, but not all, of the slope coefficients is zero.
 - (d) the slope coefficient of the variable of interest is zero, but that the other slope coefficients are not.

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Part II: Open Problems (80 points)

Multiple Regression Model (30 points)

You have data on the price and five characteristics of 399 houses in Stony Brook. The six variables are **price**: the sale price of the unit in thousands of dollars; **lotsize**: the number of square feet of the lot; **sqft**: the number of square feet in the interior of the house; **yearb**: the year the house was built; **dist**: the distance of the house to the shopping mall; and **baths**: the number of bathrooms in the house. Below is our regression output.

. reg price lotsize sqft yearb dist baths						
Source	SS	df	MS	Number of obs = 399		
Model	429305.315	5	85861.0629	F(5, 393) = 152.05		
Residual	221917.054	393	564.674437	Prob > F = 0.0000		
Total	651222.369	398	1636.23711	R-squared = 0.6592		
				Adj R-squared = 0.6549		
				Root MSE = 23.763		
price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lotsize	.0002113	.0000317	6.67	0.000	.0001491	.0002736
sqft	.0379433	.0028683	13.23	0.000	.0323043	.0435824
yearb	.5551342	.0851424	6.52	0.000	.3877426	.7225258
dist	-2.18412	.5792283	-3.77	0.000	-3.322894	-1.045347
baths	6.787188	2.346799	2.89	0.004	2.173337	11.40104
_cons	-1080.495	165.9696	-6.51	0.000	-1406.794	-754.1958

1. Write down the population model the researcher has in mind leading to the estimation of the model above. List (but do not explain) the four *OLS* assumptions made in this case.

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2. What percentage of the variation in property values is explained by the five characteristics? What is the impact of adding 1000 interior square feet on the price of a house? Explain.
3. Suppose you want to perform an “overall” significance test. Write down your null and alternative hypotheses. Perform the test at the 5% significance level based on the information in the regression output above.

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Multiple Regression Model and Prediction Intervals (30 points)

The following regression output reports *OLS* estimates of a linear model in which the price of a car depends on car characteristics like interior size, length, weight, and production origin. Production origin is represented by a dummy variable, **foreign** which equals 1 if the car is foreign and 0 if it is domestic. *Note: For any hypothesis test in this problem, even though $n = 74$, you can use the asymptotic approximation to the distribution of the test statistics.*

		Number of obs		=	74
		F(4, 69)		=	14.38
		Prob > F		=	0.0000
		R-squared		=	0.5628
		Root MSE		=	2005.9

	price	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
headroom		-481.1805	281.2525	-1.71	0.092	-1042.264	79.9028
weight		5.749148	1.500033	3.83	0.000	2.756663	8.741632
length		-81.11971	48.40356	-1.68	0.098	-177.6822	15.44275
foreign		3570.379	642.7724	5.55	0.000	2288.083	4852.675
_cons		4429.788	5035.921	0.88	0.382	-5616.6	14476.18

Test I		Test II		Test III	
test foreign headroom		test length headroom		. test length = 10*headroom	
(1) foreign	= 3600	(1) length	= 0	(1) - 10*headroom + length	= 0
(2) headroom	= 0	(2) headroom	= 0		
F(2,69) = 1.47		F(2,69) = 3.23		F(1,69) = 2.82	
Prob > F = 0.2374		Prob > F = 0.0457		Prob>F = 0.0978	

- Denote by $\hat{\beta}_0$ to the estimator of the constant, $\hat{\beta}_1$ to the estimator of the parameter of *headroom*, $\hat{\beta}_2$ to the estimator of the parameter of *weight*, and so on and so forth. Compute the *t*-statistic for $H_0 : \beta_2 = 3$ versus $H_a : \beta_2 \neq 3$. Do you reject the null hypothesis at 5% significance level? Explain.

- Construct a 90% confidence interval for β_1 .

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3. Consider Test I in the regression output. Write the null and alternative hypothesis of the test. Do you accept or reject the null at the 5% significance level? Why?

4. Suppose you have information on two different foreign cars of the same weight: A and B. Cars A and B lengths are 11 and 10 respectively while their headroom are 10 and 20 respectively. Denote the difference in price $\Delta P = P_A - P_B$ where P_A and P_B are the prices of cars A and B respectively. Rewrite the regression model so that you can perform the following test.

$$H_0 : \Delta P = 0 \text{ vs. } H_a : \Delta P \neq 0.$$

Independently of whether you could rewrite the regression or not, you can carry out the previous test using the information from Test III above. Perform the test at the 5% level.

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Linear Regression Model (20 points)

1. Explain the difference between homoskedasticity and heteroskedasticity.

2. The F -statistic with $q = 2$ restrictions when testing for the restrictions $\beta_1 = 0$ and $\beta_2 = 0$ is given by the following formula:

$$F = \frac{1}{2} \left(\frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2} \right).$$

Discuss how this formula can be understood intuitively and why do we need it.

3. What conditions does the variable Z need to satisfy in order to be considered an omitted variable? How do you use the formula

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\sigma_u}{\sigma_X} \rho_{Xu}$$

to assess the direction of the bias due to the omitted variable?

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Formulas

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_K X_{K,i} + u_i \quad (1)$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \dots - \hat{\beta}_K \bar{X}_K. \quad (2)$$

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \quad TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad SSR = \sum_{i=1}^n \hat{u}_i^2. \quad (3)$$

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\sigma_u}{\sigma_X} \rho_{Xu} \quad (4)$$

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}. \quad (5)$$

$$\bar{R}^2 = 1 - \left(\frac{n-1}{n-K-1} \right) \frac{SSR}{TSS}. \quad (6)$$

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} \quad (7)$$

$$F = \frac{1}{2} \left(\frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2} \right), \quad (8)$$

$$F = \frac{(R_{unrestricted}^2 - R_{restricted}^2)/q}{(1 - R_{unrestricted}^2)/(n - K_{unrestricted} - 1)} = \frac{(SSR_{restricted} - SSR_{unrestricted})/q}{SSR_{unrestricted}/(n - k_{unrestricted} - 1)}. \quad (9)$$

$$90\% \text{ confidence interval for } \beta_j : \left[\hat{\beta}_j \pm 1.64 \times SE(\hat{\beta}_j) \right]. \quad (10)$$

$$95\% \text{ confidence interval for } \Delta Y : \left[\widehat{\Delta Y} \pm 1.96 \times SE(\widehat{\Delta Y}) \right]. \quad (11)$$

Critical Values for 2-sided t test	
Significance	Critical value
10%	1.64
5%	1.96
1%	2.58

q	5% $F_{q,\infty}$ critical value
1	3.84
2	3.00
3	2.60
4	2.37
5	2.21